

Project 3: Open-Ended CFD Simulation with AI-Assisted Development

Present your work in the style of a journal article using scientific language and support your discussion with appropriate figures. Please use the `aapm` class inside the `REVTeX 4.2e` package as the template of your final report. The template is available on WebCampus.

Important notes before you start:

1. Your final report is **limited to 10 pages** using the required template (including references). You are required to append your code to the report; this appendix is **not** counted toward the page limit. Export or print your code as a single PDF and attach it after the references — no need to embed it in the $\LaTeX$ source.
2. This project has **two submission components**: a **Project Proposal** (see below) submitted prior to full development, and a **Final Report** submitted at the end of the semester. Both are graded.
3. Presentation really matters. Good presentation is not a subjective matter in scientific writing. You are encouraged to ask for tips and feedback at any stage. The more your document reads like an actual scientific paper, the better grade you will receive. Your interpretation and presentation of results are weighted as heavily as the results themselves.
4. **AI tools are explicitly permitted and encouraged** in this project for code generation, debugging, and exploration of new features. However, you are fully responsible for understanding, verifying, and explaining every line of code and every result you present. Uncritical use of AI-generated output without demonstrated understanding will be penalized.

Motivation

Projects 1 and 2 gave you hands-on experience implementing well-defined numerical algorithms for PDEs: a conjugate-gradient solver for elliptic equations and a fractional-step solver for the incompressible Navier–Stokes equations. In both cases, the problem, the method, and the expected deliverables were fully specified. Real research and modern engineering practice rarely works that way.

Project 3 asks you to take ownership of a CFD problem end to end: choose a problem worth solving, design a computational approach, build and verify a solver (with AI assistance where useful), run meaningful simulations, and communicate your findings as a self-contained scientific document.

Problem Statement

Choose a fluid dynamics problem that can be addressed computationally and that is meaningfully more complex or more interesting than the lid-driven cavity. Examples of suitable directions include, but are not limited to:

- Flow past bluff bodies (cylinder, airfoil, backward-facing step) at various Reynolds numbers
- Convection-driven flows: Rayleigh–Bénard convection, buoyancy-driven cavity flows, or natural convection in enclosures
- Compressible or weakly compressible effects: low-Mach flows, shock tubes
- Transport phenomena: scalar advection–diffusion, species mixing

- Extension of the lid-driven cavity solver to 3D, non-Newtonian fluids, or free-surface flows
- Turbulence modeling: RANS or large-eddy simulation of a canonical flow
- Scientific machine learning applied to CFD: physics-informed neural networks (PINNs), operator learning, or data-driven turbulence closures
- GPU or parallel acceleration of the fractional-step solver from Project 2: e.g. porting to CUDA/ROCm, using JAX or PyTorch for JIT-compiled kernels, or MPI-based domain decomposition with performance scaling analysis
- Immersed boundary methods: implement a direct-forcing or projection-based immersed boundary method to simulate flow around complex geometries on a Cartesian grid without body-fitted meshing
- Alternative spatial discretizations: re-implement a canonical flow problem (lid-driven cavity, flow past a cylinder, etc.) using finite volume methods (FVM), finite element methods (FEM), discontinuous Galerkin (DG), or spectral / spectral-element methods, and compare accuracy, convergence, and computational cost against the finite-difference baseline from Project 2

Your choice must be approved via the Project Proposal (see below). You are strongly encouraged to connect the problem to your own research interests.

Role of AI Tools

You are expected to use AI assistants (e.g. large language models such as ChatGPT, Claude, Copilot, etc.) as collaborative tools throughout the development process. The following are encouraged uses:

- Generating boilerplate code structures and debugging error messages
- Exploring numerical scheme options and asking for explanations of algorithms
- Adding new physical features or post-processing routines to an existing solver
- Drafting and refining sections of written text

The following are **not** acceptable:

- Submitting code or text you cannot explain line-by-line if asked
- Using AI output as a substitute for understanding the underlying numerics
- Fabricating or uncritically reproducing AI-generated results without independent verification

Your final report must include an AI Usage Log section (see below) documenting how AI tools were used, what prompts were effective, and critically where AI-generated output was wrong and how you caught and corrected it.

Deliverable 1: Project Proposal

Submit a **1–2 page PDF** (no specific template required, but clean typesetting is expected) addressing each of the following sections. The proposal will be evaluated on clarity, feasibility, and the degree of scientific thinking demonstrated.

1. **Problem Description** (*~1 paragraph*)
State the physical problem clearly. What flow configuration are you studying? What are the relevant governing equations and dimensionless parameters (e.g. Re , Ra , Ma)? Why is this problem scientifically or practically interesting?
2. **Numerical Approach** (*~1 paragraph*)

Describe the discretization strategy you plan to use (grid type, time-stepping scheme, solver for any linear systems). If you are extending the Project 2 solver, explain what modifications are needed. If you are starting from a different code base or using an existing open-source solver, clearly state this and justify the choice.

3. Planned Features and AI Integration

(bullet list)

List at least three specific features or capabilities you plan to implement or extend with AI assistance. For each, briefly describe what the feature is, why it is non-trivial, and how you intend to use AI tools to help build it. Examples: automated grid refinement, a new boundary condition type, a post-processing pipeline for computing force coefficients, visualization routines, etc.

4. Validation Plan

(~1 paragraph)

Identify at least one quantitative benchmark or reference dataset against which you will validate your solver. State the specific quantities you will compare (e.g. drag coefficient, Nusselt number, velocity profiles) and the reference source.

5. Expected Outcomes and Timeline

(bullet list)

List the key milestones and their target dates leading to the final submission. Be realistic about scope. A well-executed focused study is preferable to an overambitious project that is never validated.

Proposal grading (20% of Project 3 grade): clarity of problem statement (25%), quality of the numerical plan (30%), specificity of the AI integration plan (25%), and realism of the validation and timeline (20%).

Deliverable 2: Final Report

Your final report must be structured as a scientific paper using the `aapm/REVTeX 4.2e` template and must include the following sections.

1. **Abstract:** Summarize the problem, method, key results, and conclusions in 100–200 words.
2. **Introduction:** Motivate the problem, cite relevant prior work, and clearly state the objectives of your study.
3. **Governing Equations:** State all PDEs, boundary and initial conditions, and define all dimensionless parameters.
4. **Numerical Method :** Describe your discretization, time integration, and any solvers used. Include enough detail that a reader could reproduce your implementation.
5. **Verification and Validation:** Demonstrate at minimum:
 - A grid (or time-step) convergence study with observed convergence rates
 - Quantitative comparison against a published benchmark or analytical solution for at least one flow quantity
6. **Results and Discussion :** Present your simulation results with well-captioned figures. Go beyond reproducing a known benchmark — explore a parameter study, an extended case, or a physical insight that is not available from the reference alone.

7. **AI Usage Log** L A dedicated section (not counted toward page limit if it exceeds space) documenting:
 - Which AI tools were used and for what tasks
 - Representative prompts and whether the responses were useful
 - At least one concrete example where AI output was incorrect or misleading, and how you identified and corrected it
 - Your overall assessment of AI as a collaborator in CFD development
8. **Conclusion:** Summarize findings in 4–6 sentences and suggest directions for future work.
9. **References:** Cite all sources, including any open-source codes, datasets, or AI-generated content, using the `aapmrev4-2` bibliography style.
10. **Appendix: Code:** Attach your full source code as an appended PDF. The code should be well-commented.

Final report grading (80% of Project 3 grade): quality of numerical implementation and verification (30%), depth and correctness of results and discussion (30%), quality of scientific presentation (20%), and quality of AI usage log — demonstrating critical engagement, not just passive use (20%).

Note: Scope your project to be achievable within the available time. A two-dimensional flow solved rigorously and analyzed thoughtfully will earn a better grade than a three-dimensional simulation with no validation and poor discussion.